

Determination of the Matrix Free Ca^{2+} Concentration and Kinetics of Ca^{2+} Efflux in Liver and Heart Mitochondria*

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A null point titration method has been used to determine the free Ca^{2+} concentration in the matrix of rat liver and heart mitochondria which depends on addition of A-23187 to ruthenium red-inhibited mitochondria. The matrix free calcium was calculated from a knowledge of the extramitochondrial free calcium at which no net movement of calcium occurs and the ΔpH which, under these conditions, governs the equilibrium distribution of calcium across the inner mitochondrial membrane. Using this technique it was found that the relationship between total and free calcium was linear in the range studied (2–50 nmol/mg of protein). The activity coefficient for intramitochondrial calcium was 7.0×10^{-4} for liver and 4.0×10^{-4} for heart mitochondria. The apparent K_m values for matrix free Ca^{2+} were $9.7 \pm 0.5 \mu\text{M}$ for the $\text{Ca}^{2+}/2\text{H}^+$ antiporter in liver mitochondria and $5.7 \pm 0.8 \mu\text{M}$ for the $\text{Ca}^{2+}/2\text{Na}^+$ antiporter in heart mitochondria. The corresponding V_{max} values were 4.7 ± 0.4 nmol/mg of protein $\cdot$ min and 10.8 ± 2.1 nmol/mg of protein $\cdot$ min in liver and heart mitochondria, respectively. Half-maximal stimulation of the oxidation of α -ketoglutarate by heart mitochondria was observed at a matrix free Ca^{2+} concentration of about $1.2 \mu\text{M}$, which corresponds to a total calcium content of 1.5–2 nmol/mg of protein. These measurements define the permissible range whereby enzymes of the citric acid cycle may be regulated by changes of the mitochondrial Ca^{2+} concentration *in situ*.

The intramitochondrial calcium activity coefficient and the kinetic constants for Ca^{2+} efflux were measured using mitochondria isolated from livers perfused with phenylephrine or glucagon. Glucagon produced no effect on these parameters while phenylephrine caused 2-fold decreases of both the Ca^{2+} activity coefficient and the K_m for Ca^{2+} efflux. The significance of these data are discussed in relation to possible mechanisms whereby hepatic α -adrenergic stimulation may result in a net efflux of calcium from the mitochondria, thereby accounting for the rise of cytosolic free Ca^{2+} .

Mitochondria as normally isolated from a variety of organs contain between 5 and 15 nmol/mg of protein of total calcium. Despite a great deal of work in the area, it has proved a difficult and unresolved problem to determine the relationships between bound and free Ca^{2+} in the mitochondrial matrix and the concentration gradient of free Ca^{2+} across the

mitochondrial membrane (1, 2). It is now generally accepted that in energized mitochondria Ca^{2+} uptake and efflux occurs by separate and independent pathways (1–4). Calcium uptake is mediated by an electrophoretic process, with transport occurring down the proton electrochemical gradient generated by the mitochondrial electron transport oxidation-reduction proton pumps. As already emphasized by a number of investigators (5, 6), if the Ca^{2+} distribution across the mitochondrial membrane achieves thermodynamic equilibrium with the electrical component of the normal proton electrochemical gradient according to the Nernst equation, a Ca^{2+} gradient of 10^4 to 10^5 is expected. For a cytosolic free Ca^{2+} concentration of $0.2 \mu\text{M}$ (7), a calcium gradient of this magnitude would correspond to a mitochondrial matrix free Ca^{2+} concentration of 2 to 20 mM. At this Ca^{2+} concentration, the solubility product for calcium phosphate would be exceeded. Deposits of calcium phosphate have not been observed in mitochondria except under severe pathological conditions, suggesting that the free Ca^{2+} concentration is very much lower. From measurements of the extramitochondrial free Ca^{2+} and the membrane potential, as the latter was varied by Ca^{2+} loading or by K^+ plus valinomycin addition, calculations of the matrix free Ca^{2+} from the Nernst equation provided values of 0.1 to 0.01 for the calcium activity coefficient in liver (8) and heart (9) mitochondria. For mitochondria containing 10 nmol of Ca^{2+} per mg of protein, this corresponds to a matrix free Ca^{2+} concentration of 0.1 mM to 1 mM and a Ca^{2+} concentration gradient of 500 to 5000.

A different approach to the estimation of mitochondrial free Ca^{2+} has been used by Denton and colleagues (10–13) and by Hansford (14, 15), based on measurements of the activity of various mitochondrial Ca^{2+} -sensitive enzymes as a function of the Ca^{2+} concentration calculated from Ca^{2+} -EGTA¹ buffers. α -Ketoglutarate dehydrogenase, NAD-linked isocitrate dehydrogenase, and pyruvate dehydrogenase phosphatase assayed in crude mitochondrial extracts were each found to be activated by Ca^{2+} , with half-maximum effects obtained at about $1 \mu\text{M}$ free Ca^{2+} (10). Titrations of α -ketoglutarate and pyruvate dehydrogenase activities against extramitochondrial free Ca^{2+} concentration using mitochondria from heart or adipose tissue showed that the free Ca^{2+} concentration required to produce half-maximum enzyme activation varied from about 40 nM under state 3 conditions to about $1 \mu\text{M}$ in the uncoupled state (11). If the Ca^{2+} sensitivity of the enzymes in the crude extracts is assumed to be the same as in the intact mitochondria, these data suggest that under energized conditions mitochondria maintained a Ca^{2+} concentration gradient of only

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¹ The abbreviations used are: EGTA, ethylene glycol bis(β -aminoethyl ether)- N,N,N',N' -tetraacetic acid; HEPES, 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid; EDTA, ethylenediaminetetraacetate; MOPS, 4-morpholinepropanesulfonic acid; DMO, 5,5-dimethylloxazolidine-2,4-dione; FCCP, carbonyl cyanide *p*-trifluoromethoxyphenylhydrazone.

about 25 across the mitochondrial membrane. The intramitochondrial calcium content was not measured under the conditions of these experiments, but this Ca^{2+} gradient at an extramitochondrial free Ca^{2+} concentration of $0.2 \mu\text{M}$ and a total mitochondrial calcium content of 10 nmol/mg of protein would correspond to a matrix free Ca^{2+} concentration of $5 \mu\text{M}$ and an activity coefficient of 5×10^{-4} , assuming a matrix volume of $1 \mu\text{l/mg}$ of protein.

In a previous communication from this laboratory we described a null point titration technique for the estimation of the mitochondrial free Ca^{2+} and the matrix calcium activity coefficient (16). This method made use of the calcium ionophore A-23187 to equilibrate Ca^{2+} across the mitochondrial membrane while ruthenium red was added to inhibit the Ca^{2+} uniporter. In these experiments, the pH gradient across the mitochondrial membrane was thought to be collapsed in the presence of A-23187, and it was assumed that the measured calcium null point of $64 \mu\text{M}$ for normally prepared rat liver mitochondria was equal to the matrix free Ca^{2+} concentration. By varying the mitochondrial total calcium content, a calcium activity coefficient² of 35×10^{-4} was estimated. Values for the mitochondrial free Ca^{2+} in the range of $50\text{--}100 \mu\text{M}$ suggested that the calcium-sensitive enzymes in the mitochondria would always be saturated with calcium (17). In view of the discrepancy between our previous study and the work of Denton and McCormack (reviewed in Ref. 17), we have re-evaluated our previous methodology and report our new findings in this paper. In addition, we have used the mitochondrial null point titration technique to determine the kinetics of Ca^{2+} efflux in rat liver and heart mitochondria and the effect of phenylephrine treatment of perfused rat liver on the kinetics of Ca^{2+} efflux of subsequently isolated mitochondria. Preliminary accounts of parts of this work have been published (18, 19).

EXPERIMENTAL PROCEDURES

Preparation of Mitochondria—Rat liver mitochondria were prepared by slight modifications of standard techniques (20). The isolation medium consisted of 225 mM mannitol, 75 mM sucrose, 5 mM HEPES, and 0.1 mM EDTA, pH 7.2. The final wash was carried out in the same medium from which EDTA had been omitted. The mitochondria were then resuspended in 185 mM mannitol, 75 mM sucrose, and 20 mM MOPS, pH 7.2, which had been passed several times through a Chelex-100 column to decrease the free Ca^{2+} concentration to below $1 \mu\text{M}$. Rat heart mitochondria were prepared essentially as described by LaNoue *et al.* (21) using the same isolation medium as for liver mitochondria. The finely minced tissue from four hearts was mixed with 20 ml of isolation medium containing 1 mg/ml of Nagarse, and homogenized for 2 min , diluted 4-fold with isolation medium, and homogenized for a further 1 min . Heart mitochondria were subsequently isolated using the same techniques as for liver mitochondria. Respiratory control ratios with 10 mM glutamate and 1 mM malate as substrate were 4 to 6 for heart mitochondria and at least 7 for liver mitochondria.

Assay for Free Ca^{2+} —Uptake or release of mitochondrial calcium was determined spectrophotometrically using antipyrilazo III (bis(4-antipyrilazo)-4,5-dihydroxy-2,7-naphthalene disulfonic acid) as a metallochromic Ca^{2+} indicator at a wavelength pair of $720\text{--}790 \text{ nm}$ (16, 22). Unless otherwise indicated, mitochondria were incubated at 30°C , pH 7.2, in basic media containing 140 mM KCl, 20 mM MOPS, and $100 \mu\text{M}$ antipyrilazo III. When present, substrate concentrations were: 10 mM succinate plus $0.5 \mu\text{M}$ rotenone or 10 mM ascorbate plus 0.5 mM N,N,N',N' -tetramethyl-*p*-phenylenediamine. Because of the

nonlinearity of the change in absorbance of the Ca^{2+} -antipyrilazo III complex over the calcium concentration range from zero to $300 \mu\text{M}$ (22), the absorbance changes at different Ca^{2+} concentrations were calibrated by the further addition of small amounts of calcium as internal standard. The free Ca^{2+} concentration in the medium for a given total calcium concentration was calculated from a knowledge of the total antipyrilazo III concentration and the apparent dissociation constant (K'_d) for the Ca^{2+} -antipyrilazo III complex. The K'_d values for the Ca^{2+} -antipyrilazo III complex were determined for each different incubation medium, as described previously for the Ca^{2+} -arsenazo III dissociation constant (7), using an absolute extinction coefficient for antipyrilazo III of 30 mM^{-1} . The apparent calcium dissociation constant for antipyrilazo III in the basic incubation medium was $155 \mu\text{M}$ (*cf.* Ref. 22). Over the range of calcium concentrations used in this study, 70–80% of the total calcium was in the free form with $100 \mu\text{M}$ antipyrilazo III.

Measurement of $\Delta\psi$, ΔpH , and Total Calcium—The electrical potential $\Delta\psi$ across the mitochondrial membrane was determined from the distribution of [^3H]methyltriphenylphosphonium bromide ($3 \mu\text{Ci/ml}$) essentially as described previously (23). The pH gradient and matrix water spaces were also measured according to techniques previously used in this laboratory (23) with the exception that the matrix pH was measured both from the distribution of [^{14}C]DMO ($1 \mu\text{Ci/ml}$) or [^{14}C]acetate ($1 \mu\text{Ci/ml}$). The water space was measured with $^3\text{H}_2\text{O}$, which was subtracted from the [^{14}C]sucrose space ($1 \mu\text{Ci/ml}$) to give the matrix water space. The total calcium content of mitochondria was either measured spectrophotometrically with antipyrilazo III by addition of A-23187 in the presence of ruthenium red ($1 \mu\text{M}$) or by atomic absorption. For the latter technique 0.05 ml of the concentrated mitochondrial suspension containing approximately 40 mg of protein/ ml was acidified with 0.1 ml of 14% (w/v) perchloric acid, sonicated for 5 s , and centrifuged for 30 s in an Eppendorf centrifuge. An aliquot of the supernatant (0.1 ml) was diluted to 1 ml with a mixture of strontium chloride and potassium chloride for calcium assay by atomic absorption spectroscopy. The mitochondrial calcium content was calculated after subtraction of blank values obtained by using the same procedure with 0.05 ml of the suspension medium. Mitochondrial calcium loading was achieved as previously described (24). Protein was assayed by the biuret method using crystalline bovine serum albumin as standard (25).

Mitochondrial Calcium Depletion and Assay of α -Ketoglutarate Dehydrogenase—Rat heart mitochondria as normally prepared contained 5 to 8 nmol of calcium/ mg of protein. In order to decrease the endogenous calcium content to between 1 and 2 nmol of calcium/ mg of protein, rat heart mitochondria (about 20 mg of protein/ ml) were incubated for 10 min at room temperature in medium containing 120 mM KCl, 20 mM HEPES, 0.1 mM MgCl_2 , and 0.5 mM EGTA, pH 7.2. After centrifugation, the mitochondria were resuspended in mannitol-sucrose-MOPS medium containing $20 \mu\text{M}$ EGTA. α -Ketoglutarate dehydrogenase activity in intact rat heart mitochondria was measured polarographically by monitoring the rate of α -ketoglutarate oxidation (11). Rates of oxygen consumption were measured at 30°C by incubating rat heart mitochondria (1 mg of protein/ ml) in Chelex-treated medium containing 120 mM KCl, 20 mM K^+ -HEPES, 5 mM phosphate-Tris, 1 mM K^+ -malate, and 2.5 mM ADP, pH 7.2, to which was added consecutively 0.5 mM and 20 mM α -ketoglutarate. Enzyme activity is expressed as the ratio of rates obtained with 0.5 mM and 20 mM α -ketoglutarate after correction for the basal rate observed prior to α -ketoglutarate addition (11).

RESULTS

Mitochondrial Calcium Null Point—The method for estimating the free Ca^{2+} concentration in the matrix space of isolated mitochondria relies on the fact that the ionophore A-23187 forms a complex with Ca^{2+} , which is transported across the mitochondrial membrane by an electroneutral exchange of Ca^{2+} with 2H^+ (26, 27). Thus, when mitochondria are incubated in a suitable buffer in the presence of a Ca^{2+} indicator, addition of A-23187, in the presence of ruthenium red to inhibit the Ca^{2+} -uniporter, will cause either a release or uptake of Ca^{2+} dependent on the initial Ca^{2+} concentration gradient. When the extramitochondrial free Ca^{2+} is such that no net movement of Ca^{2+} across the mitochondrial membrane occurs upon addition of A-23187, the extramitochondrial concentrations of free Ca^{2+} [Ca^{2+}]_o and H^+ [H^+]_o will be related

² In this context the mitochondrial activity coefficient is defined as the ratio of the matrix free Ca^{2+} concentration to the bound Ca^{2+} concentration, with the latter being calculated by dividing the total calcium content by the matrix water volume. This use of the term "activity coefficient" is different from that relating concentration and thermodynamic activity of Ca^{2+} in solution due to high ionic strength effects. Our data have not been corrected for such deviations from ideal behavior.

to the intramitochondrial concentrations of Ca^{2+} $[\text{Ca}^{2+}]_i$, and H^+ $[\text{H}^+]$, according to the relationship

$$[\text{Ca}^{2+}]_o/[\text{Ca}^{2+}]_i = 10^{2\Delta\text{pH}} \quad (1)$$

Under these conditions, referred to as the Ca^{2+} null point, the matrix free Ca^{2+} concentration $[\text{Ca}^{2+}]_i$ can be calculated if the proton gradient is known.

Fig. 1 gives a series of representative traces showing net Ca^{2+} changes after addition of 0.4 nmol of A-23187/mg of protein to a suspension of rat liver mitochondria incubated with 1 μM ruthenium red, 0.14 nmol of nigericin/mg of protein, and different initial extramitochondrial Ca^{2+} concentrations. Calcium changes were measured using 100 μM antipyrilazo III as a spectrophotometric calcium indicator, and the free Ca^{2+} concentration was calculated using the Ca^{2+} -antipyrilazo dissociation constant measured in the same manner. Release of Ca^{2+} was observed with an extramitochondrial free Ca^{2+} of 87 μM and below, while uptake of Ca^{2+} was seen with 200 μM free Ca^{2+} . The small overshoot of Ca^{2+} release after A-23187 addition when the extramitochondrial free Ca^{2+} approached the null point (e.g. 87 μM Ca^{2+}) might be caused by a transient disequilibrium between Ca^{2+} and H^+ movements since A-23187 will also equilibrate Mg^{2+} across the mitochondrial membrane. At each Ca^{2+} concentration, calibration was achieved by addition of known amounts of calcium. The amount of calcium uptake or release from the mitochondria from the time they were added to the cuvette to completion of the A-23187-induced change is then plotted as a function of the initial free Ca^{2+} concentration of the buffer to obtain a Ca^{2+} null point plot.

Results from a similar experiment are shown plotted in Fig. 2A. The line is extrapolated to the abscissa by linear regression analysis, and the extramitochondrial free Ca^{2+} concentration for zero net Ca^{2+} movement across the mitochondrial membrane in the presence of A-23187 gives the mitochondrial

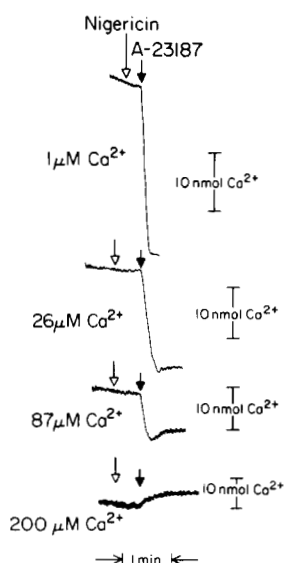


FIG. 1. Spectrophotometric traces to illustrate the method used to determine the calcium null point. Liver mitochondria (1.2 mg/ml) were incubated in 2.5 ml of a medium containing 140 mM KCl, 20 mM MOPS, 100 μM antipyrilazo III, and 0.5 μM ruthenium red in the absence of any added calcium (upper trace) or in the presence of various concentrations of calcium as indicated. Nigericin (0.14 nmol/mg of protein) and A-23187 (0.4 nmol/mg of protein) were then added. The change in calcium concentration observed in the medium was calculated using the absorbance change (720–790 nm) obtained on addition of a known amount of a standardized solution of calcium. A downward deflection of the trace represents Ca^{2+} release from the mitochondria. In this experiment the control mitochondrial calcium content was 15.5 nmol/mg of protein.

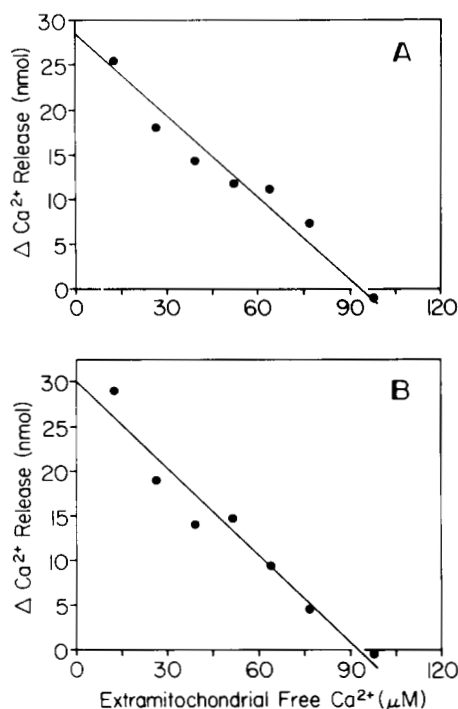


FIG. 2. A comparison of the null points obtained with calcium release measured either spectrophotometrically or by atomic absorption spectroscopy. Two aliquots of liver mitochondria (8 mg of protein) were incubated in cuvettes containing 3 ml of the basic medium (see legend to Fig. 1) and placed in the thermostatted cuvette housing (30 °C) of a dual wavelength spectrophotometer. After 10 s of incubation, a 1-ml sample was removed from the cuvette not under observation and centrifuged at $10,000 \times g$ in an Eppendorf centrifuge. Nigericin (0.14 nmol/mg of protein) and A-23187 (0.40 nmol/mg of protein) were added to both incubations, and another 1-ml sample was removed from the sample cuvette when the spectrophotometric trace indicated the completion of calcium movements. The calcium release observed at various calcium concentrations, calculated from the spectrophotometric traces (see Fig. 1), is shown in A. The net changes in calcium, calculated as the mean of changes observed in the mitochondrial pellet and supernatant as measured by atomic absorption spectroscopy, are shown in B. The initial mitochondrial calcium content was 10.7 nmol/mg of protein.

calcium null point for a particular total calcium content. For comparison, Fig. 2B gives a similar calcium null point titration with the same batch of mitochondria but with net calcium changes being measured by atomic absorption. As seen from the figure, agreement of the calcium null point values is excellent despite different systematic errors between the two methods for measuring the calcium changes.

Under similar conditions of incubation, the mitochondrial Ca^{2+} null point is dependent on the initial mitochondrial calcium content. Fig. 3 illustrates that a series of parallel lines are produced for Ca^{2+} null point titrations using mitochondria containing initial calcium contents from 6.4 to 16.3 nmol/mg of protein.

Mitochondrial Free Ca^{2+} Concentration—Knowledge of the pH gradient across the mitochondrial membrane under conditions of the Ca^{2+} null point titrations is required in order to estimate the matrix free Ca^{2+} concentration from the measured Ca^{2+} null point. As shown in Table I, the distribution of $[\text{H}^+]$ acetate (pK of 4.64) and $[\text{H}^+]$ DMO (pK of 6.32) is compared in parallel incubations in order to obtain a more accurate value for the mean ΔpH . Under all conditions the two indicators gave very similar values. Under control conditions the ΔpH was 0.9 unit and the membrane potential ($\Delta\psi$) was 148 mV, giving an overall value of about 202 mV for the proton electrochemical gradient ($\Delta\mu\text{H}^+$). Addition of nigericin

caused a partial collapse of the pH gradient, which was compensated by an increase of $\Delta\psi$, while addition of A-23187 alone produced similar changes. Little further changes were produced by the combined presence of nigericin and A-23187. Thus, in the high KCl medium used, the residual ΔpH in the presence of nigericin and A-23187 was 0.46 unit. The last line of Table I shows that a concentration of FCCP sufficient to cause a maximal stimulation of respiration and collapse of the $\Delta\psi$ to 40 mV decreased ΔpH to 0.31 unit. A complete collapse or even inversion of the ΔpH has been observed only with mitochondria incubated in low K^+ medium (28). The fact that the distributions of two weak acids of differing pKs gave essentially the same value for the matrix pH under different conditions suggests that possible isotope binding artifacts cannot account for the apparent persistence of a residual ΔpH across the mitochondrial membrane even in the presence of an uncoupling agent. The absence of significant binding of the weak acids to the mitochondria was confirmed by finding no acetate or DMO gradients between the medium and detergent-treated mitochondria. Additional studies showed that the residual ΔpH value of 0.46 measured in the presence of nigericin and A-23187 was independent of the calcium content of the mitochondria up to values of at least 50 nmol/mg of protein.

It is evident from Equation 1 that the matrix free Ca^{2+} calculated from the measured Ca^{2+} null point is very dependent on the accuracy of the ΔpH measurement. As a further check on the validity of the measurements, Ca^{2+} null points were determined in the presence of A-23187 and nigericin and compared with those obtained with the same batch of mitochondria incubated with A-23187 and FCCP. Under the latter conditions, the ΔpH is 0.15 unit less than in the standard

incubation buffer, and hence for the same matrix free Ca^{2+} concentration, the Ca^{2+} null point should be diminished by half (antilog 0.3). The data shown in Fig. 4 confirm this prediction since the Ca^{2+} null point was 104 μM in the presence of A-23187 plus nigericin and 48 μM in the presence of A-23187 and uncoupler, yielding values of 12.5 μM and 11.5 μM , respectively, for the matrix free Ca^{2+} concentration. In another experiment where there was a 0.1 unit ΔpH difference between control and uncoupled conditions, the calculated matrix free Ca^{2+} was 9.2 μM and 10.0 μM , respectively. Thus, agreement between the paired values is within 8%.

Mitochondrial Calcium Activity Coefficient—Liver mitochondrial Ca^{2+} null points were determined at a series of different mitochondrial calcium contents, as illustrated in Fig. 3, and the matrix free Ca^{2+} concentrations were calculated using Equation 1. A plot of mitochondrial calcium content against the mitochondrial free Ca^{2+} concentration for a series of experiments with liver mitochondria is shown in Fig. 5A. The relationship was linear up to about 50 nmol of calcium per mg of protein, while the ratio of total to free calcium was 0.95 ± 0.02 nmol/mg of protein $\cdot \mu\text{M}$. The results of a similar plot with heart mitochondria are shown in Fig. 5B, which gave a ratio of total to free calcium of 1.37 ± 0.09 nmol/mg of protein $\cdot \mu\text{M}$. Heart mitochondria incubated in the presence of nigericin and A-23187 under conditions of the calcium null point measurements had a residual ΔpH of 0.49 ± 0.04 and a matrix space of 0.52 ± 0.05 $\mu\text{l}/\text{mg}$ of protein (5 determinations in each group). Using matrix water spaces of 0.67 $\mu\text{l}/\text{mg}$ of protein for liver mitochondria and 0.52 $\mu\text{l}/\text{mg}$ of protein for heart mitochondria, the calcium activity coefficients² for liver and heart mitochondria are 7.0×10^{-4} and 4.0×10^{-4} , respectively. Further experiments with liver mitochondria showed

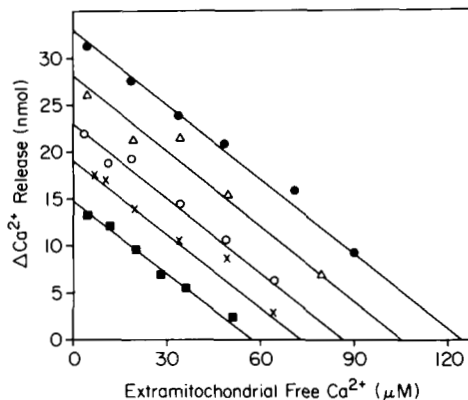


FIG. 3. Dependence of the null point on the initial calcium content of the mitochondria. Calcium null point titrations were carried out with liver mitochondria (0.8 mg of protein/ml) as described in Fig. 1. The initial calcium contents (nmol/mg of protein) in this experiment were: ■—■, 6.4; ×—×, 9.2; ○—○, 10.2; △—△, 12.9; ●—●, 16.3, as determined by atomic absorption spectroscopy.

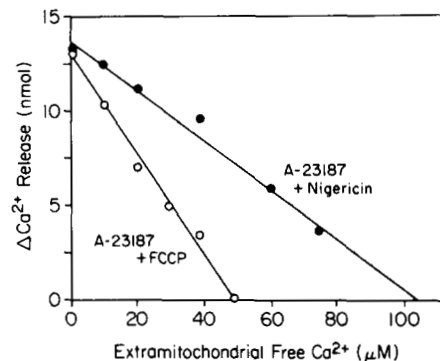


FIG. 4. The effect of changes in the mitochondrial ΔpH on the measured calcium null point. Liver mitochondria (0.55 mg of protein/ml) were incubated in the basic incubation medium, and a null point titration was performed as described in Fig. 1 (●—●) or with FCCP (0.4 μM) replacing nigericin (○—○). Parallel incubations were carried out for the determination of ΔpH and matrix volumes as described under "Experimental Procedures."

TABLE I
Effects of A-23187, nigericin, and FCCP on ΔpH and $\Delta\psi$ in rat liver mitochondria

Rat liver mitochondria (5 mg/ml) were incubated for 3 min in basic medium containing in addition 6% (w/v) dextran and 1 μM ruthenium red, pH 7.2. The concentration of ionophores was as follows: nigericin, 0.14 nmol/mg of protein; A-23187, 0.4 nmol/mg of protein; and FCCP, 0.5 nmol/mg of protein. Values shown are means $\pm$ S.E. of 4–8 experiments, each with triplicate determinations.

Additions	ΔpH DMO	ΔpH acetate	ΔpH mean	Matrix space $\mu\text{l}/\text{mg}$	$\Delta\psi$ mV	$\Delta\mu\text{H}^+$
Control	0.92 ± 0.01	0.89 ± 0.03	0.90 ± 0.03	0.87 ± 0.03	148.4 ± 1.2	201.7
Nigericin	0.55 ± 0.02	0.46 ± 0.04	0.50 ± 0.03	0.62 ± 0.02	171.0 ± 1.3	200.6
A-23187	0.56 ± 0.02	0.52 ± 0.04	0.53 ± 0.02	0.74 ± 0.04	172.6 ± 1.6	204.0
A-23187 + nigericin	0.46 ± 0.02	0.46 ± 0.03	0.46 ± 0.02	0.67 ± 0.03	169.2 ± 1.3	196.4
FCCP	0.30 ± 0.03	0.32 ± 0.01	0.31 ± 0.02	0.92 ± 0.03	39.8 ± 2.8	58.2

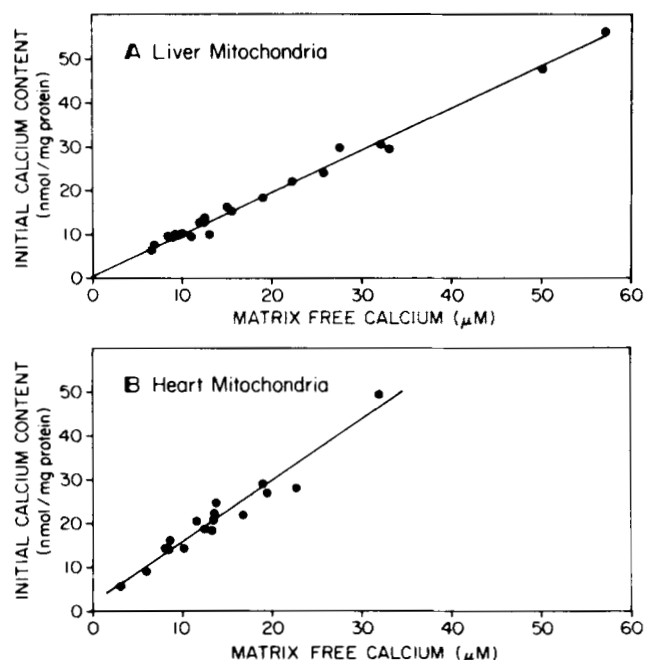


FIG. 5. The relationship between free and total calcium in rat liver and heart mitochondria. Calcium null point titrations were carried out with liver and heart mitochondria containing different initial calcium contents. The matrix free calcium corresponding to these null points was calculated after correction for the residual ΔpH (see text). The points shown for both liver (A) and heart (B) mitochondria are the combined data obtained in experiments with 5 separate preparations of mitochondria.

that addition of 1 mM Mg^{2+} increased the calcium activity coefficient by 25%, while a decrease of pH to 6.8 in the presence of 1 mM Mg^{2+} had a similar effect. In other experiments, the presence of 10 mM acetate had no effect on the apparent calcium activity coefficient at pH 7.2 (data not shown). Replacement of KCl by sucrose or choline chloride in the mitochondrial incubation medium also had no effect on the measured calcium activity coefficient.

Kinetics of Ca^{2+} Efflux from Mitochondria—Ruthenium red at low concentrations selectively inhibits Ca^{2+} uptake into mitochondria by the Ca^{2+} -uniporter without affecting the Ca^{2+} efflux by electroneutral exchange with H^+ or Na^+ (29, 30). Since the Ca^{2+} null point titration method permits the calculation of the mitochondrial free Ca^{2+} concentration at different levels of calcium loading, it becomes possible to investigate the kinetics of the ruthenium red-insensitive pathway of Ca^{2+} efflux from mitochondria. Fig. 6A shows a Lineweaver-Burk plot for Ca^{2+} efflux from liver mitochondria incubated at pH 7.2 with basic KCl medium in the absence of Mg^{2+} . Averaged data from eight experiments provided mean values $\pm$ S.E. of 4.7 ± 0.4 nmol/min·mg of protein for the apparent V_{max} and 9.7 ± 0.5 μM for the apparent K_m of Ca^{2+} efflux for liver mitochondria. Ruthenium red-insensitive calcium efflux from heart mitochondria is very slow in the absence of Na^+ (30); hence for the experiments represented in Fig. 6B for heart mitochondria, the incubation medium contained 10 mM Na^+ but was otherwise similar to that used for liver mitochondria. Means $\pm$ S.E. for five experiments gave values of 10.8 ± 2.1 nmol/min·mg of protein for the apparent V_{max} and 5.7 ± 0.8 μM for the apparent K_m of Na^+ -activated Ca^{2+} efflux for heart mitochondria. Further experiments showed that 4 mM Na^+ gave a half-maximal stimulation of Ca^{2+} efflux in heart mitochondria, in agreement with earlier studies (30).

Calcium Sensitivity of α -Ketoglutarate Oxidation by Rat Heart Mitochondria—An important application of the Ca^{2+}

null point titration method for measuring the mitochondrial matrix free Ca^{2+} concentration relates to the determination of Ca^{2+} sensitivity of mitochondrial Ca^{2+} -responsive enzymes. Hansford and Castro (15), using intact rat heart mitochondria, have shown that a 50% activation of α -ketoglutarate dehydrogenase was achieved with a mitochondrial calcium content of about 1 nmol/mg of protein while full activation was obtained with an endogenous calcium content of 3–4 nmol/mg of protein. In the above study, α -ketoglutarate dehydrogenase activity in isolated heart mitochondria was estimated from the steady state level of reduction of endogenous pyridine nucleotides after addition of 0.5 mM α -ketoglutarate plus 0.5 mM malate to mitochondria containing different amounts of total calcium and incubated in the absence of ADP. In Fig. 7 the pooled results of similar experiments are shown except that α -ketoglutarate dehydrogenase activity was estimated from the ratio of the rates of respiration obtained at low (0.5 mM) and high (20 mM) α -ketoglutarate concentrations in the presence of ADP (11). The total mitochondrial calcium content (upper scale of Fig. 7) was related to the matrix free Ca^{2+} concentration (lower scale of Fig. 7) by the relationship depicted in Fig. 5B. Fig. 7 shows that a half-maximal stimulation of α -ketoglutarate dehydrogenase activity was obtained at a matrix free Ca^{2+} concentration of about 1.2 μM .

Hormonal Effects on Mitochondrial Ca^{2+} Binding and Efflux Kinetics—A further application of the method presented in this paper is an assessment of the effects of hormone pretreatment of perfused rat livers on the matrix free Ca^{2+} concentration and the kinetics of the ruthenium red-insensitive Ca^{2+} efflux pathway in subsequently isolated mitochondria. Fig. 8 shows the relationships between total mitochon-

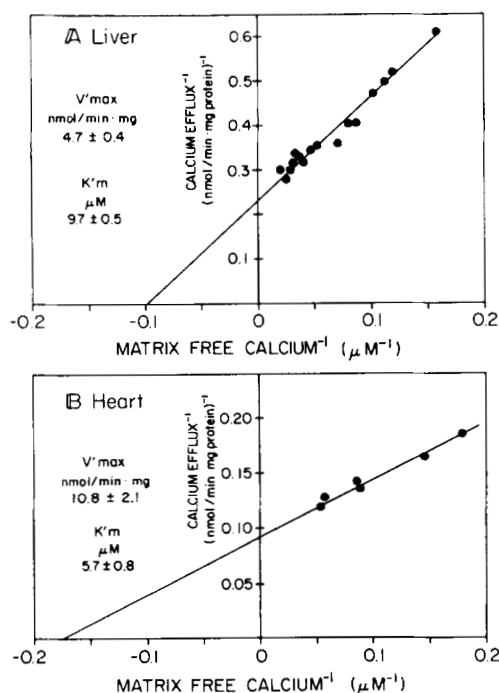


FIG. 6. Kinetics of calcium efflux in rat liver and heart mitochondria. The initial rate of calcium efflux on addition of ruthenium red (0.5 μM) was measured in rat liver (A) and heart (B) mitochondria at different calcium loadings. The matrix free calcium corresponding to the initial calcium content was calculated from the activity coefficient determined as in Fig. 5. The data are presented in the form of a Lineweaver-Burk plot and are from representative experiments. The kinetic parameters given in the figures are the mean $\pm$ S.E. of the values obtained in separate experiments carried out on five liver and three heart mitochondrial preparations.

drial calcium and the matrix free Ca^{2+} concentration in mitochondria isolated from livers of fed rats perfused for 2 min with 10^{-5} M phenylephrine or perfused for 5 min with 10^{-7} M glucagon prior to homogenization, compared with mitochondria isolated from control perfused liver. Glucagon pretreat-

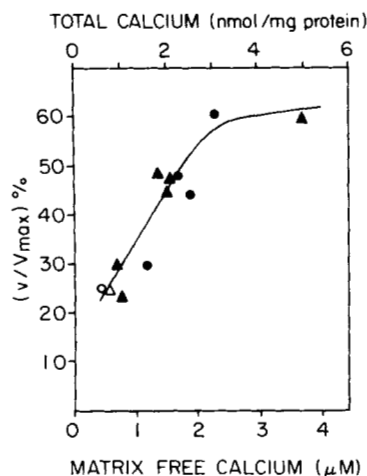


FIG. 7. The dependence of α -ketoglutarate oxidation in rat heart mitochondria on the intramitochondrial free calcium concentration. α -Ketoglutarate oxidation in the presence of ADP (2.5 mM) was measured in calcium-depleted rat heart mitochondria. The intramitochondrial calcium content was varied by the addition of small amounts of calcium up to a maximum of 7 nmol/mg of protein. The lowest calcium contents were obtained by the addition of 10 mM NaCl + 8 μ M EGTA (Δ , $\circ$). The intramitochondrial calcium content was measured in parallel incubations performed in KCl medium (see "Experimental Procedures") from which ADP had been omitted. A 1-ml aliquot was transferred from the electrode chamber into a cuvette containing 1.5 ml of medium supplemented with 140 μ M Arsenazo III and 1 μ M ruthenium red at the same time as addition of 0.5 mM α -ketoglutarate (3 min). The total calcium released from the mitochondria on addition of A-23187 (0.4 nmol/mg of protein) was calculated from the absorbance increase measured at 675–685 nm using an internal calcium standard. The symbols (Δ , $\bullet$) represent data points from two mitochondrial preparations. The matrix free Ca^{2+} concentration corresponding to the total calcium content was calculated from the data in Fig. 5B.

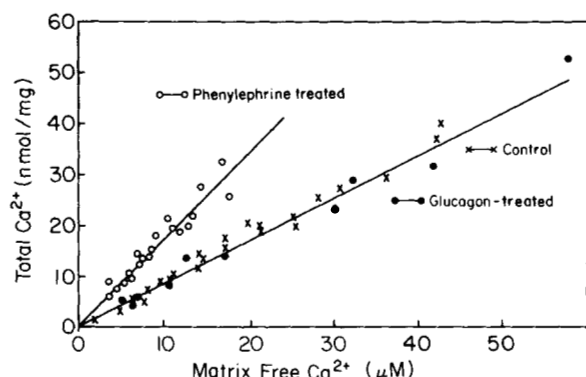


FIG. 8. The relationship between total and free calcium in mitochondria isolated from livers perfused with glucagon and phenylephrine. Livers from fed rats were perfused, in a nonrecirculating mode, with Krebs-Henseleit bicarbonate buffer containing 1.25 mM CaCl_2 for 15 min. Mitochondria were then isolated as described under "Experimental Procedures." Where livers were treated with hormones, the mitochondria were isolated after a further 2-min perfusion with phenylephrine (10^{-5} M) or a further 5-min perfusion with glucagon (10^{-7} M). Null point titrations were then carried out on mitochondria isolated from control ($\times$ — $\times$), phenylephrine ($\circ$ — $\circ$) or glucagon ($\bullet$ — $\bullet$) treated livers at various initial calcium contents as described in Fig. 3. The results presented are the combined data obtained in mitochondria prepared from 8 control, 5 phenylephrine, and 4 glucagon-treated livers.

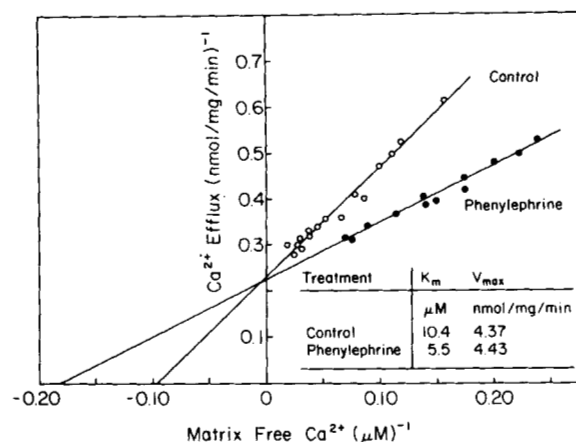


FIG. 9. Effect of phenylephrine treatment on the kinetics of calcium efflux in rat liver mitochondria. Mitochondria were isolated from livers perfused in the presence or absence of phenylephrine as described in the legend to Fig. 8. The initial rate of calcium efflux was measured, at different intramitochondrial calcium contents, in the presence of 0.5 μ M ruthenium red (see Fig. 6). The matrix free calcium concentration corresponding to the total calcium content in mitochondria from control and phenylephrine-treated livers was obtained from Fig. 8. A Lineweaver-Burk plot of the combined data from 8 control and 5 hormone-treated livers is shown.

ment had no effect on the ratio of total to free Ca^{2+} in subsequently isolated mitochondria or on the total calcium content in paired experiments. However, phenylephrine pretreatment caused a decrease of total mitochondrial calcium, as previously observed (7), and a 2-fold increase of total to free Ca^{2+} in subsequently isolated mitochondria. This latter result implies that interaction of phenylephrine with the α_1 -receptors of the liver plasma membrane resulted in a rapid increase in the affinity of the bulk binding sites in the mitochondria for calcium. Further experiments showed that this effect could not be attributed to changes of the mitochondrial Mg^{2+} content or the free Mg^{2+} concentration after phenylephrine treatment of livers since there was no effect on these parameters.³ It has also been shown that phenylephrine addition to isolated hepatocytes does not affect the cytosolic-mitochondrial ΔpH (31). In all the above experiments the responsiveness of the perfused livers to hormone additions was ascertained from measurements of the rate of glucose release.

Fig. 9 shows the effect of 2 min of phenylephrine pretreatment of livers on the kinetics of ruthenium red-insensitive Ca^{2+} efflux from subsequently isolated mitochondria. These experiments are analogous to those shown in Fig. 6A. The hormone treatment had no effect on the apparent V_{max} of Ca^{2+} efflux, but the apparent K_m was decreased from 10.4 μ M to 5.5 μ M. Glucagon treatment, on the other hand, had no effect on the kinetics of ruthenium red-insensitive Ca^{2+} efflux from liver mitochondria (data not shown).

DISCUSSION

A number of indirect methods have been used previously to determine the activity coefficient of Ca^{2+} in the intramitochondrial compartment. A value of 0.1–1.0 was obtained by Azzone *et al.* (5) using a method based on the Ca^{2+} -induced swelling of mitochondria in the presence of acetate. EPR spectroscopy, using Mn^{2+} as an analogue of Ca^{2+} , yielded values of 0.017, in the absence of acetate, and 0.32 when 50 mM acetate was present (32, 33). If the distribution of calcium across the mitochondrial membrane was governed solely by Ca^{2+} transport on an electrogenic uniporter, then the equilib-

³ B. E. Corkey and J. R. Williamson, unpublished experiments.

rium distribution ratio of Ca^{2+} activities would be related to the mitochondrial membrane potential by the Nernst equation. Measurements of the membrane potential and Ca^{2+} distribution ratio should then permit the determination of the calcium activity coefficient. However, this approach is valid only if the gradient of free Ca^{2+} is in equilibrium with the membrane potential. This condition was thought to be met at high intramitochondrial calcium content, low membrane potential, and in the presence of a permeant anion (34). Using this method, the activity coefficient for Ca^{2+} has been reported to be 0.1 in rat liver (8) and 0.025–0.044 in rat heart mitochondria (9). These values are all much higher than the measured calcium activity coefficient of 7×10^{-4} and 4×10^{-4} obtained in the present study for liver and heart mitochondria, respectively. Consequently, it appears that more than 99.9% of the total mitochondrial calcium is always bound and that the Ca^{2+} gradient is never in equilibrium with the membrane potential.

It is of interest to consider the nature of the intramitochondrial sites that are responsible for the binding of calcium. The relationship between total and free calcium in both rat heart and liver mitochondria was linear up to a calcium content of 50 nmol/mg of protein. This implies the presence of a large number of binding sites. Since no attempt was made in this study to saturate these sites, an equilibrium binding analysis was not possible. Nevertheless, inspection of Fig. 5 would indicate that the dissociation constant of the binding sites must be greater than 50 μM . Calcium is known to interact with negatively charged phospholipids, and their mitochondrial content and affinity for calcium is in the range which suggests that they could play a major role in Ca^{2+} binding (35–37). The Ca^{2+} activity coefficient was found to be lower in heart than in liver mitochondria. The different phospholipid compositions of the inner mitochondrial membranes may account for this difference (38, 39). In the null point titration method A-23187 is employed to equilibrate Ca^{2+} across the mitochondrial membrane. This ionophore efficiently transports Mg^{2+} as well as Ca^{2+} (26, 27). If Mg^{2+} and Ca^{2+} competed for the same intramitochondrial binding sites, a lower matrix free Ca^{2+} concentration would be expected in Mg^{2+} depleted than in Mg^{2+} loaded mitochondria. In fact, only a small change in matrix free calcium is observed when the null point titrations are carried out in 1 mM MgCl_2 , which corresponds to the Mg^{2+} null point, suggesting that Mg^{2+} does not compete effectively for the calcium-binding sites. A similar conclusion is reached for the possible interaction of H^+ with the mitochondrial matrix Ca^{2+} -binding sites. Further studies are clearly required to determine the nature of these sites and their affinity for Ca^{2+} under different conditions. Similar values for the Ca^{2+} activity coefficient were obtained whether the mitochondria were suspended in KCl, sucrose, or choline chloride. However, values for the mitochondrial Ca^{2+} activity coefficient reported in this paper should be regarded as condition-dependent, and extrapolations to other mitochondrial conditions, or species, should be made with caution.

From a knowledge of the activity coefficient for Ca^{2+} , the kinetic parameters of various intramitochondrial Ca^{2+} -dependent processes can be determined. One of these is the kinetics of Ca^{2+} efflux from the mitochondria. Considerable evidence is available to indicate that mitochondrial Ca^{2+} influx and efflux occur by separate carriers (1–4). Calcium influx occurs via an electrophoretic uniporter which is inhibited by ruthenium red, while Ca^{2+} efflux occurs by a ruthenium red-insensitive electroneutral exchange of Ca^{2+} with either H^+ or Na^+ , depending on the mitochondrial species. Thus, when mitochondria are incubated under steady state conditions, Ca^{2+} cycles across the mitochondrial membrane at a rate determined by the kinetic parameters of the Ca^{2+} uptake and

efflux pathways so that the extramitochondrial free Ca^{2+} concentration achieves an equilibrium with the mitochondrial matrix calcium (4). Perturbation of the extramitochondrial Ca^{2+} concentration under these conditions by addition of Ca^{2+} or EGTA results in a net Ca^{2+} uptake or release until the original steady state Ca^{2+} concentration of 0.3 to 1 μM is restored (8, 40, 41). The value of 4.7 nmol/min·mg of protein measured for the V_{max} of Ca^{2+} efflux in this paper agrees well with other measurements made with calcium-loaded mitochondria (8) and is several hundred-fold lower than the V_{max} for Ca^{2+} uptake (42, 43). Our findings that the ratio of total to free Ca^{2+} in the matrix of liver mitochondria is approximately 1 nmol/mg· μM and the K_m for Ca^{2+} efflux on the $\text{Ca}^{2+}/2\text{H}^+$ antiporter is 9.7 μM accounts for the observation of Nicholls (8) that in liver mitochondria containing above 10 nmol of calcium/mg of protein the Ca^{2+} efflux pathway is close to saturation. Under these conditions, the equilibrium value for the extramitochondrial free Ca^{2+} concentration is relatively insensitive to increases of the mitochondrial calcium content, and the mitochondria have been considered to buffer the extramitochondrial Ca^{2+} (4). Under equilibrium conditions of Ca^{2+} cycling, Ca^{2+} influx becomes kinetically limited by the extramitochondrial free Ca^{2+} concentration, which is 10- to 20-fold lower than the K_m of the Ca^{2+} uniporter. As illustrated by a kinetic model of mitochondrial Ca^{2+} cycling (2), when the matrix free Ca^{2+} concentration falls below the K_m for the Ca^{2+} efflux antiporter, the rate of Ca^{2+} efflux is diminished and is matched by a lower rate of Ca^{2+} influx as a consequence of a fall in the equilibrium value of the extramitochondrial free Ca^{2+} concentration. Under these conditions, addition of a small amount of Ca^{2+} to mitochondria, sufficient to increase the total calcium content by a few nanomoles per mg of protein causes significant increases in both the intramitochondrial free Ca^{2+} concentration and the equilibrium value for the extramitochondrial free Ca^{2+} concentrations.⁴ Thus, when mitochondria contain low amounts of calcium, such that the free Ca^{2+} concentration is below the K_m for the efflux pathway, the intramitochondrial free Ca^{2+} concentration is relatively sensitive to changes of the extramitochondrial free Ca^{2+} concentration. Denton and McCormack (13) have shown that α -ketoglutarate dehydrogenase as well as NAD-linked isocitrate dehydrogenase and pyruvate dehydrogenase phosphatase in mitochondrial extracts from different organs are activated by Ca^{2+} with a $K_{0.5}$ of about 1 μM and have suggested that the activity of these enzymes in the intact cell should be influenced by changes of the cytosolic free Ca^{2+} concentration. The data shown in Fig. 7 for the effect of changes in the matrix free and total Ca^{2+} on the activity of α -ketoglutarate dehydrogenase in intact heart mitochondria illustrate that the enzyme activity increases over the range of total calcium between 0.5 to 3 nmol/mg of protein and of free Ca^{2+} up to 2.5 μM . Consequently, the activity of α -ketoglutarate dehydrogenase is likely to be regulated by changes of the cytosolic free Ca^{2+} only if the calcium content of mitochondria in the intact heart cell is below 3 nmol/mg of mitochondrial protein. At present there is inadequate data to assess whether or not this is so.

As a further application of the null point titration method for measuring the mitochondrial free Ca^{2+} concentration we have also investigated the effects on mitochondria of glucagon and phenylephrine administration to perfused rat livers. The activity coefficient for Ca^{2+} was measured in mitochondria isolated from hormone-treated livers. Glucagon (10^{-7} M) had no effect on this parameter (Fig. 8), which is in accordance

⁴ S. K. Joseph, K. E. Coll, and J. R. Williamson, unpublished observations.

with evidence indicating that glucagon has no acute effects on hepatic calcium homeostasis (reviewed in Ref. 2). On the other hand, stimulation of α_1 -adrenergic receptors in liver is known to cause an increase of the cytosolic free calcium concentration within 2 min (7) and an increase of phosphorylase activity (see Ref. 44 for review), presumably as a consequence of Ca^{2+} -calmodulin activation of phosphorylase kinase. In previous studies we (7) and others (45, 46) have presented evidence showing that the calcium content of the mitochondrial fraction of the cell decreased shortly after addition of phenylephrine to hepatocytes or perfused liver. This finding suggested that the mitochondria provided the precursor calcium pool for the elevated cytosolic Ca^{2+} concentration, at least for hepatocytes incubated in Ca^{2+} -free media. One explanation advanced to account for the findings was that an α_1 -adrenergic second messenger produced a direct effect on mitochondrial calcium homeostasis, such that Ca^{2+} efflux was stimulated relative to Ca^{2+} influx (7, 46). Measurements of the activity coefficient for calcium in mitochondria isolated from livers perfused for 2 min with phenylephrine (Fig. 8) showed a 2-fold decrease over control, suggesting a generalized increase in the affinity of the intramitochondrial binding sites for Ca^{2+} . Thus, in the mitochondria from phenylephrine-treated livers, there was not only a fall of the total calcium content but also a 2-fold decrease of free Ca^{2+} at any given level of total calcium over the measured range from 2 to 18 nmol/mg of protein. Since the hormone effect on the apparent activity coefficient of mitochondrial calcium is rapid, it cannot be associated with any major changes of phospholipid content or composition but would require more subtle changes, perhaps relating to the detailed environment of certain phospholipid head groups within the bilayer structure of the membrane.

Measurements of the initial rate of Ca^{2+} efflux in ruthenium red-inhibited mitochondria as a function of the matrix free Ca^{2+} concentration showed that there was approximately a 2-fold decrease of the apparent K_m without any change of $V_{\max}$ in the mitochondria from phenylephrine-treated livers (Fig. 9). However, it is not clear whether this represents a specific hormone-mediated effect on the $\text{Ca}^{2+}/2\text{H}^+$ antiporter or whether the lowered K_m reflects the generalized increased affinity of Ca^{2+} binding to all Ca^{2+} -binding sites as noted above. In any case, the combination of a decrease in both the K_m for Ca^{2+} efflux and the free Ca^{2+} concentration in the mitochondrial matrix without a change of $V_{\max}$ is insufficient to account for enhanced mitochondrial Ca^{2+} efflux in phenylephrine-stimulated livers, as illustrated by the model of mitochondrial Ca^{2+} cycling (2). On the other hand, an increase in the K_m of the Ca^{2+} uniporter would be accompanied by an increase of the extramitochondrial free Ca^{2+} concentration and a decreased mitochondrial calcium content. Mg^{2+} is known to increase the apparent K_m for Ca^{2+} influx on the Ca^{2+} uniporter (47-49), and an elevation of the extramitochondrial Mg^{2+} concentration has been demonstrated to cause an increase in the equilibrium value of extramitochondrial free Ca^{2+} (8, 40). However, there is no evidence that the phenylephrine effect on intracellular Ca^{2+} redistribution in liver is caused by changes of the cytosolic free Mg^{2+} concentration, since our preliminary data (50) show that phenylephrine addition to hepatocytes actually decreased the cytosolic free Mg^{2+} concentration slightly from 0.37 ± 0.04 mM to 0.24 ± 0.02 mM (mean of 4 observations). Furthermore, our attempts to determine directly whether the apparent K_m for Ca^{2+} influx is increased in mitochondria isolated from phenylephrine-treated livers have so far been unsuccessful. In conclusion, although we have demonstrated an effect of phenylephrine on the matrix Ca^{2+} -binding properties of subsequently isolated

liver mitochondria, the significance of this observation with regard to α -adrenergic alterations of hepatic calcium homeostasis is presently not clear.

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